



Tribhuvan University
Faculties of Humanities and Social Sciences

A Project Report
On
STUDENT MANAGEMENT SYSTEM

Submitted to
Department of Computer Application
Universal College
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1. Introduction

Student Management System (SMS) is a fully computerized system where all the student related data can be stored, retrieved, monitored & analyzed. The data is saved at a centralized location & role-based login access is given to all the stakeholders for ensuring student data security. It improves data accuracy, enhance communication, and increase efficiency in managing student information

Our 'Student Management System' aids in managing, storing, data such as admission number, tracking attendance, exam & test marks, notice board, teacher info, subject info and other student related data. The system gives both the student and teacher an overall view of the ongoing academic activities.

2. Problem Statement:

The common issue faced by student management in educational institutions is the data loss, manual monitoring of students' performance and etc. Students form the main part of any institution that concerns with. But the institution finds it difficult to keep the details of all the students. Thus, it involves a lot of paperwork. Sometimes there will be huge heap of files bundled up and must be stored in some corner of the office. Managing student records manually is a troublesome job, this increase as the number of students' increases. This method may lead to missing or destroying important information and is also difficult to maintain. Since this method of managing student data is outdated, we propose a better way, i.e., using computer software which can automate most of the work in maintaining such records.

3. Objectives:

The primary objectives of the student management system are as follows:

- To a secure and reliable platform to store and manage student data.
- To enable teachers and administrators to record and monitor student performance.

4.Methodology

a. Requirement Identification

i. Study of existing system

School currently use a manual system (handwritten) for the management of student data such as admission number, tracking attendance, exam & test marks, notice board, teacher info, subject info and other student related data. Due to this the common issue faced by student management in educational institutions is the data loss, manual monitoring of students' performance and etc. Students form the main part of any institution that concerns with. But the institution finds it difficult to keep the details of all the students. Thus, it involves a lot of paperwork. Sometimes there will be huge heap of files bundled up and must be stored in some corner of the office. Managing student records manually is a troublesome job, this increase as the number of students' increases.

ii. Requirement collection

A computerized way of handling information about property and users' details is efficient, organized and time saving, compared to a manual way of doing so. This is done through a database driven web application whose requirements are mentioned in this section.

Software Requirement

Database Support : MySQL 8.4 or above

Operating system : Windows 10/11 or Mac 14/15 Operating System

Web Browser : Google Chrome 120 or later

Coding language : PHP 8.4, HTML5, CSS3, JS (ES6+).

Server Deployment: Apache Server, MySQL Server.

Hardware Requirement

CPU : Intel Core i3 / AMD Ryzen 3 or above.

Memory (Primary): 4 GB or above.

Hard Disk : 20 GB available SSD space or above.

Monitor : 15" Full HD Color Monitor. Keyboard and Mouse.

b. Feasibility Study

i. Technical Feasibility

The technical feasibility study aims to assess the practicality and viability of developing and implementing a student management system (SDMS) from a technical standpoint. It involves evaluating the system's requirements, technology stack, infrastructure, and resources. Here are key considerations:

- ❖ **System Requirements:** Evaluate the specific requirements of the SDMS, including data storage capacity, performance expectations, scalability, and integration needs. Ensure that the proposed system can meet these requirements effectively.
- ❖ **Infrastructure:** Evaluate the necessary infrastructure components, such as servers, networking, and storage systems, to host and support the SDMS. Determine whether the existing infrastructure is sufficient or if additional resources or upgrades are required.
- ❖ **Integration Capabilities:** Analyze the integration requirements with other existing systems or databases within the educational institution. Assess the compatibility and feasibility of integrating the SDMS with these systems, ensuring smooth data exchange and interoperability.
- ❖ **Development Resources:** Evaluate the availability of skilled developers and technical personnel who can handle the development, implementation, and maintenance of the SDMS. Consider the required expertise in database management, user interface development, system administration, and security.
- ❖ **Scalability and Performance:** Assess the system's ability to handle increasing data volumes, concurrent users, and future growth. Consider factors such as database optimization, caching mechanisms, load balancing, and the use of scalable technologies to ensure the system can scale effectively.
- ❖ **Security and Data Privacy:** Evaluate the measures required to ensure the security and privacy of student data within the SDMS. Consider encryption, access controls, secure authentication mechanisms, and regular security audits to mitigate potential risks.
- ❖ **Technical Support and Maintenance:** Determine the availability of technical support and maintenance resources to address issues, apply updates, and provide ongoing

support for the SDMS. Consider the need for documentation, user training, and a helpdesk or support system.

❖ **Technology**

- HTML is used for the front-end design. It provides a means to structure text-based information in a document. It allows users to produce web pages that include text, graphics and hyperlinks.
- CSS (Cascading Style Sheets) is a style sheet language used for describing the presentation of a document written in a mark-up language. Although most often used to set the visual style of web pages and user interfaces written in HTML, the language can be applied to any XML document.
- SQL is the language used to manipulate relational databases. It is tied closely with the relational model. It is issued for the purpose of data definition and data manipulation.
- PHP is a server-side scripting language that is embedded in HTML. It is used to manage dynamic content, databases, and session tracking. PHP Syntax is C-Like.
- JavaScript (JS) is a scripting language, primarily used on the Web. It is used to enhance HTML pages and is commonly found embedded in HTML code.

ii. **Economical Feasibility**

- The economical feasibility contains the documentation regarding
- the cost and benefits of new proposed computerized system.

iii. **Operational Feasibility**

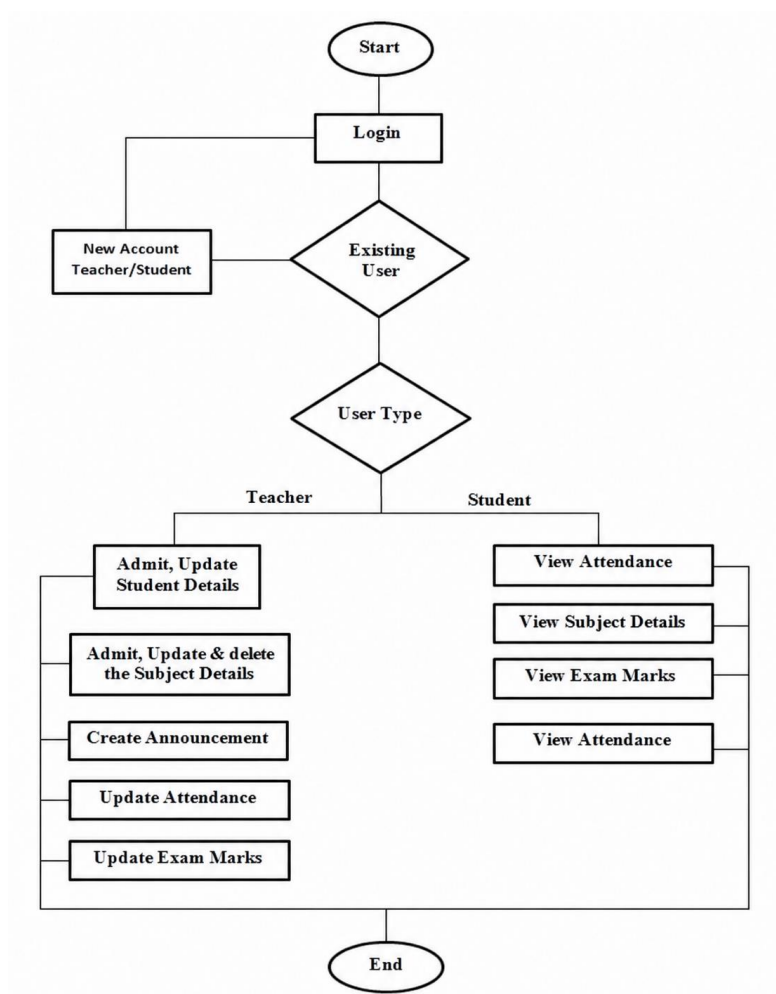
- User Acceptance: Evaluate the willingness of users (administrators, teachers, staff, and students) to adopt and use the student management system. Conduct surveys or interviews to gather feedback and understand their expectations, concerns, and preferences.
- Training and Support: Assess the feasibility of providing adequate training and support to users for effectively utilizing the system. Determine the availability of resources, such as training materials, user guides, and technical support personnel, to assist users during the implementation and ongoing usage of the system.
- Change Management: Evaluate the feasibility of managing the organizational change associated with implementing the student management system. Identify potential resistance to change and develop strategies to mitigate it. Assess the

readiness of the educational institution to embrace the new system and adapt their processes accordingly.

- **Process Integration:** Analyze the feasibility of integrating the student management system into existing operational processes and workflows. Evaluate the impact on existing procedures, roles, and responsibilities, and identify any necessary modifications or adjustments to ensure smooth integration.

c. High-Level Design System

i. Flowchart



5.Gantt chart (project Timeline)

GANTT CHART FOR STUDENT MANAGEMENT SYSTEM

Activities	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
Requirement Gathering								
System Analysis								
Database Design								
Development								
Testing								
Documentation								
Final Submission								

6.Expected Outcomes

The expected outcome of implementing a student management system is a secure and reliable platform to store and manage student data, ensuring easy accessibility. It automates administrative tasks such as enrolment, attendance tracking, and grade management, reducing manual effort and saving time for staff members. The system enhances accessibility and allowing for secure and controlled access to up-to-date student records. Real-time data analysis and reporting capabilities enable data-driven decision-making and proactive identification of student needs. With robust security measures in place, the system ensures data privacy and compliance with regulations. Overall, the expected outcome is an improved educational administration process, effective communication, and better utilization of data for informed decision-making.

7.References

- [1] Bregzis, R., Gotlieb, C., & Moore, C. (2002). The beginning of automation in the University of Toronto Library, 1963-1972. *IEEE Annals of the History of Computing*, 24(2), 50-70.
- [2] Desai, S. K., Attar, S., Mane, S. H., Dethe, K. B., & Lomte, A. (2020). Online Scientists Information System using AES. *International Journal of Computer Applications*, 176(11), 29-31.
- [3] Fu, X., Peltsverger, B., Qian, K., Tao, L., & Liu, J. (2008). APOGEE: automated project grading and instant feedback system for web based computing. *ACM SIGCSE Bulletin*, 40(1), 77-81.
- [4] Jou, M., Shiau, J. K., & Zhang, H. W. (2009). Application of web technologies in automation technology education. *International Journal of Computers and Applications*, 31(4), 215-221.
- [5] Langde, S., Maurya, A., Nakhawa, T., Sinha, A., Patil, S., Karanam, K., & Muguttrao, H. (2018). Automated Attendance System. *International Journal of Applied Research*, 248-249.